Image downloads: what is tried, in what order, and the fallbacks

Solar_data_mining-main | Multiwb_datamine.get_SDO_imgs / get_LASCO_imgs / get_STEREO_imgs | state of the code on 2026-11-10

Step | Decision | Result saved / returned | Fallback or gap (logged, no crash) | Helioviewer JP2

SDO: get_SDO_imgs (AIA, HMI)

1. Jobs
One job per (instrument, product), e.g. (AIA,193), (HMI,magnetogram). nproc=1 serial, else a multiprocess Pool across jobs.

2. Search + download (find_coobs)
Fido/VSO (JSOC) search at a.Sample(dT); HMI filtered by product. Files go to outdir/<INSTR>/<wave>/raw/.

3. Prepare
AIA: Level 1 -> aiapy update_pointing + register -> Level 1.5 (0.6"/px, DN/s, north-up, PC = identity), file rewritten as *lev15*.
HMI: kept as delivered (pixels unrotated, CROTA2 ~ 180 deg; header is truthful).

on error

If search/download fails
Warning; the job returns what it has (possibly an empty list). Nothing stops the other jobs.

4. Select per slot (_series)
All FITS in the product folder; the image nearest each cadence slot sT, sT+dT, ... eT is kept (duplicates dropped).

5. Save
_save_maps -> outdir/<INSTR>/<wave>/<INSTR>_<wave>_<time>.fits (full WCS, unchanged roll). PNG quicklooks.

6. Helioviewer back-up (helioviewer=True)
A JP2 for **every** cadence slot -> raw/helioviewer/ (hv<id>_<time>.jp2). If the nearest JP2 falls outside the window, one retry nudged inward. JP2s are not returned.

Return
List of FITS Map lists, one per job, in input order.

SOHO: get_LASCO_imgs (C2, C3)

1. need_calib = True? Route: Level-1 -> Helioviewer -> Level-0.5
Yes: VSO/SDAC search with a.Level(1); nearest record per slot; kept only if LEVEL >= 1 and BUNIT = MSB. Kept files saved **unchanged** in maps/ (NRL already rotated them north-up).

none usable

2. No usable Level-1 -> Helioviewer
A JP2 per cadence slot (cache reused; one inward retry at the window edges) -> FITS in **maps_HV/**: canonical tags (OV_SRC Helioviewer JP2, BUNIT ct), roll 0 (OV_ROLL kept), SOHO observer, occulter masked. For 2025-05-30 VSO returned only Level-0.5, so this is used.

fails

3. Helioviewer also fails -> last fallback
Archive Level-0.5 route below (DN/s), with a warning.

Archive route (need_calib=False, or last fallback)
VSO search padded +/-30 min; if VSO fails, the LASCO team archive index (img_hdr.txt). Only standard 1024x1024 frames (C2 orange / C3 clear). Nearest record per slot; a failing file is replaced by the next nearest (up to 4 rounds). need_calib=False with helioviewer=True also saves a JP2 for every slot.

4. Prepare (_prepare_lasco)
Divide by EXPTIME -> DN/s; mask occulter and outside the FOV (C2 2.4-6, C3 3.8-30 Rsun). Pixels unrotated; spacecraft CROTA (~7 deg) kept; SOHO observer added.

5. Save + return
Level-1 and Level-0.5 in maps/, Helioviewer in maps_HV/; PNG previews. Returns the Maps.

STEREO: get_STEREO_imgs (EUVI, COR1, COR2)

1. Jobs
One job per (instrument, wave): EUVI x waves, COR1, COR2. Pool across jobs if nproc>1.

2. Reuse local FITS (_stereo_series)
Read FITS already in STEREO_<sc>/<instr>[/<wave>]/. Filters: spacecraft, detector, EUVI wavelength, COR1 polarizer = polar, **COR2 POLAR = 1001** (total brightness, d*c2 files). A slot is covered if a match lies within dT/2. No SECCHI Level-1 is distributed, so consumers use the route Helioviewer -> Level-0.5.

3. Uncovered slots only
One Fido/VSO search (SECCHI, provider SSC data archive). Per slot, download candidates nearest-first (within dT/2) until one passes the filters. Single-polarizer n*c2 files are downloaded but rejected.

none

No match for a slot
Logged "Missing STEREO_<sc>/<instr>: time" (no crash).

4. Save
maps/STEREO_<sc>_<instr>[_wave]_<time>.fits **unchanged**: Level-0.5 secchi_rectify, DN, roll in the PC matrix (~6.5-6.9 deg), observer HGLN/HGLT/DSUN kept. No secchi_prep, no exposure normalisation. retrieval_log.txt.

5. Helioviewer back-up (helioviewer=True)
A JP2 for every slot -> raw/helioviewer/ (header roll already 0, pixels north-up, observer kept).

Return
One Map list per job. Browse JPGs (stereo-ssc .../browse/) are never used.

After download: one product family per analysis

Route everywhere: Level-1 -> Helioviewer -> Level-0.5 (LASCO when need_calib=True; STEREO always). Science FITS live in **maps/**, Helioviewer FITS in **maps_HV/** (or the overlay maker's maps/*_HV_*.fits), JP2s in **raw/helioviewer*/**. Every JP2-to-FITS conversion writes identical tags, so all Helioviewer FITS form one family. The flow code uses the first family in the route that covers every cadence slot, otherwise the one covering most slots (ties by the route), and never mixes families in a difference. **get_helioviewer_jp2** and **jp2_to_fits** are public.

Spacecraft roll: how orientation is set for every source

Rule: every saved FITS header truthfully describes its pixels; every consumer loads through load_map and only follows the WCS. Nothing rotates an image because of its source or level.

What each product looks like

Product	Pixels	Header roll	What happens on load (load_map)
LASCO Level-0.5 / DN/s (archive)	as observed	CROTA ~ 7 deg (C2 ~ 7.7)	unchanged; SunPy applies the roll
LASCO Level-1 (NRL)	north-up	matches pixels	unchanged; no further edits
AIA Level 1.5 (aiapy)	north-up	PC = identity	unchanged
HMI (JSOC)	as observed	CROTA2 ~ 180 deg	unchanged; SunPy applies it
SECCHI EUVI/COR1/COR2 L0.5	as observed	PC matrix ~ 6.5-6.9 deg	unchanged; observer kept
Helioviewer JP2 (all)	north-up	LASCO: original CROTA; others: 0	SunPy reads 0 for LASCO only via a JP2-only key
maps_HV FITS (jp2_to_fits)	north-up	0 (old in OV_ROLL)	unchanged
Old Helioviewer FITS (before 2026-11-10)	north-up	stale CROTA ~ 7 deg	**corrected to 0 in memory**

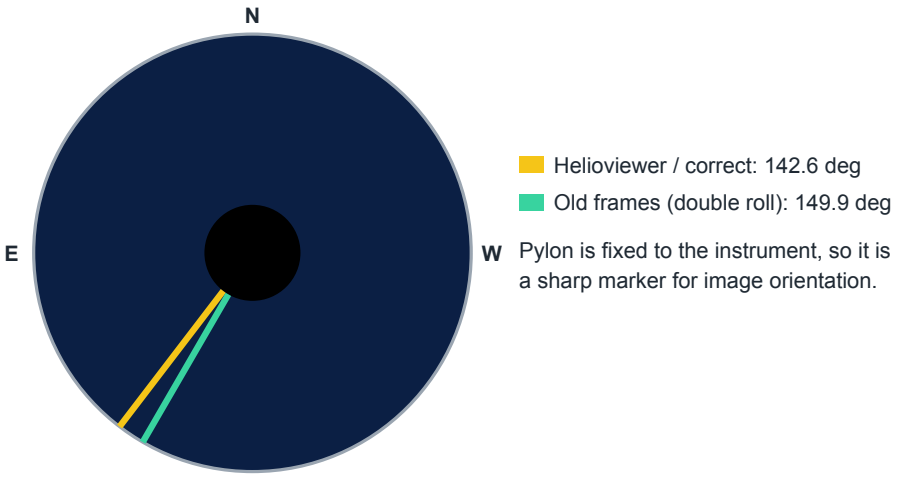
The double-roll problem (found 2026-11-10)

SunPy ignores the CROTA of a Helioviewer LASCO JP2 only while its *helioviewer* header key exists. Writing the image to FITS dropped that key, so reloading the FITS rotated the already north-up pixels again: about 7 deg counter-clockwise for C3 and 7.7 deg for C2. The frame and flow code then showed C3 structures ~7 deg away from the PFSS lines (PFSS itself is placed from sky coordinates and was right). Affected: all Helioviewer-derived LASCO FITS (maps/, difference FITS) made before the fix.

Where the roll is set in the code

- **_hv_north_up(meta)**: CROTA/CROTA1/CROTA2 -> 0, PC -> identity, old value -> OV_ROLL.
- **_get_lasco_source** (Helioviewer branch) and **jp2_to_fits** call it before saving.
- **_save_maps**: any Helioviewer-derived Map is saved with the rotation SunPy used in memory.
- **load_map**: a Helioviewer-derived image with a nonzero roll and no OV_ROLL gets roll 0 in memory; everything else is returned as it is. Used by solar_overlay, both frame makers and the flow code.
- **_prepare_lasco**: verified Level-1 MSB returned untouched; otherwise DN/s + masks, roll untouched.
- Consumers rotate only via the WCS: SunPy rotate() / reproject when the rotation matrix is not identity.

Check: C3 pylon position angle (counter-clockwise from north)



Measurement (2025-05-30/31 event)	Result	Meaning
C3 pylon: Helioviewer JP2 / your screenshot	142.6 / 142.4 deg	reference, north-up
C3 pylon: raw Level-0.5 rotated by its own CROTA	143.3 deg	archive header is right
C3 pylon: old frame reference grid	149.9 deg	double roll (+7 deg CCW)
C3 pylon: new flow cache from the same old files	142.9 deg	load_map fixes it
C3 pylon: maps_HV (jp2_to_fits) / DN/s rotated	142.6 / 143.2 deg	families agree
C2 Helioviewer maps: best-fit roll vs header	0.0 vs 7.7 deg	same fault in C2
Best-fit roll of Helioviewer JP2: HMI, AIA 193, EUVI 195, COR2	0.0 deg each	JP2 headers right

Using the images safely

- Load every image with **load_map**; never re-rotate by source or level.
- Difference only within one family: science FITS with science FITS, Helioviewer with Helioviewer (the overlay and flow code refuse mixed pairs).
- Helioviewer pixels are display values (0-255): fine for positions, morphology and apparent motion, not for brightness or mass.
- Old Helioviewer FITS on disk still carry the stale roll; consumers fix it on load. ~/Documents/Codes/fix_helioviewer_lasco_roll.py repairs the headers themselves (dry run by default).
- Frames made before 2026-11-10 from Helioviewer LASCO data are rotated ~7 deg (C3) / 7.7 deg (C2) and should be regenerated.